

Histogram, Contrast Adjustment

The histogram of an image represents the distribution of grayscale. It gives an estimate of the probability density of light intensity. It is the basic tool for setting acquisition parameters (gain, offset, gamma), improving the contrast of the image, or choosing a good binarization threshold.

- 1) Write a program to read and display a grayscale image. Display its histogram. What do the X and Y axes represent? Observe the histogram of various images at your disposal: **pout.tif**, **cameraman.tif**, **bureau.gif**, **rice.tif**, **tire.tif**, **circuit.tif**, **ic.tif**, ...
- 2) Histogram stretch: Display the minimum and maximum grayscale of and then apply an affine transformation to match the minimum grayscale to 0 and the maximum grayscale to 1 (or 255). What does this transformation produce on the contrast of the image?
- 3) Gain/Offset Setting: Apply an affine transformation of the intensity: $O = Gain * I + Offset$

Investigate the influence of gain and offset adjustment on multiple images. How do these two parameters act on the histogram? Which histogram should be looked for when acquiring an image to make full use of the sensor's dynamic range and achieve good contrast?

- 4) Gamma correction: corresponds to a non-linear transformation of the intensity

$$O = \left(\frac{I}{255} \right)^\gamma \times 255$$

What brightness correction causes a value $\gamma < 1$? a value $\gamma > 1$? Correct an underexposed image.

See: https://docs.opencv.org/3.4/d3/dc1/tutorial_basic_linear_transform.html

- 5) Histogram equalization: this transformation is calculated to distribute the intensities as well as possible, using the entire range uniformly (i.e. to approach a flat histogram). Apply histogram equalization to different images under different light conditions and observe the results.

See: https://docs.opencv.org/4.x/d5/daf/tutorial_py_histogram_equalization.html